Dimitris Tsitsigkos

Software Engineer (PhD)

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Work Experience

Postdoctoral researcher (Data Science and Engineering team) Feb 2025 – Mar 2026

Archimedes Research on AI, Data Science and Algorithms, Athens, Greece

- Designed and implemented high-performance, in-memory multi-dimensional and vector indices in C++, using parallelism and hardware-aware optimizations to improve query throughput and latency
- Technologies: **C++, OpenMP, SIMD**

PhD fellow Jan 2024 – Dec 2024

Department of Computer Science & Engineering, Univ. of Ioannina, Greece

- Contributed to the implementation of spatial index structures in C++ for distance queries (k-NN, range, distance joins), improving query performance compared to existing approaches
- Contributed to the design and development of a prototype distributed spatial data management system using MPI and OpenMP, reducing memory consumption by over 55% while improving query performance.
- Technologies: **C++, OpenMP, MPI**

PhD fellow Jul 2019 – Dec 2024

Information Management Systems Institute, Athena Research Center, Athens, Greece

- PhD Research:
  - Optimized B+-Tree Index Structures
    - Implemented BS-tree, a gapped data-parallel B+-tree, and its compressed variant CBS-tree for high-performance in-memory indexing
    - Both BS-tree and CBS-tree achieved 1.5×–2.5× higher throughput versus B+-trees, learned indexes, and trie-based approaches across different read and update workloads
    - CBS-tree uses 56%–94% less memory than competing indexes on compressible datasets while maintaining superior throughput
    - Designed SIMD-friendly node layouts, branchless search, and efficient gap handling to fully exploit data-parallel CPU features, boosting single-threaded performance and scaling across cores
  - Indexing and Partitioning for Non-Point Spatial Data
    - Developed two-layer space-oriented partitioning for indexing non-point spatial objects
    - Reduced spatial join cost by ∼50% compared to existing spatial indexing methods
    - Achieved 2–3× faster range query performance vs R-trees, quad-trees and grid-based indexes
    - Designed techniques supporting parallel spatial query processing
  - Parallel Spatial Join Processing
    - Implemented a parallel spatial join algorithm for rectangles using the Partition-Based Spatial Merge (PBSM)
    - Optimized 1D/2D partitioning, sweep strategies, and data-driven parameter tuning to maximize join performance
    - Achieved parallel spatial joins of tens of millions of rectangles in <1 s on multi-core systems, demonstrating high scalability and efficiency
- Participated in European funding project:
  - MORE: Management of Real-time Energy data.
    - Built parallel continuous evaluation module in Java for sliding-window aggregations, enabling low-latency energy analytics
    - Implemented parallel versions of time-series analytics algorithms in Python using Dask to improve efficiency and scalability for large-scale data processing
- Technologies: **C++, OpenMP, SIMD, Python, Dask**

Software Engineer Apr 2017 – Nov 2018

Hellenic Army Information Technology Support Center, Athens, Greece

- Maintained and extended Java backend applications for internal systems, improving performance and implementing new features
- Technologies: **Java, Oracle Database, JSF**.

Research Engineer Dec 2012 – Jun 2019

Information Management Systems Institute, Athena Research Center, Athens, Greece

Participated in several Greek and European funding projects, including:

- Amnesia: A platform for anonymizing relational, multi-dimensional, and hierarchical data.
  - Implemented an open-source data anonymization platform for in-memory anonymization, supporting both web and desktop applications
  - Designed the architecture and full stack, with a Java backend providing RESTful APIs (Spring) and a web frontend using JavaScript and HTML
  - Integrated the platform into the OpenAIRE ecosystem, enabling global researchers, data providers, and companies to anonymize data efficiently
- MoDisSENSE: A Distributed Spatio-Temporal and Textual Processing Platform for Social.
  - Built distributed spatio-temporal processing (Java/Hadoop/HBase) for trajectory reconstruction and POI analytics with RESTful APIs and PostgreSQL
- Technologies: **Java, RESTful Web Services, Spring, MapReduce, Hadoop, HBase, PostgreSQL**

Technical Skills

Languages:	C++, Java, Python
Data Management:	MySQL, PostgreSQL, PostGIS, HBase
Parallel & Distributed Systems:	OpenMP, SIMD, Spark, MPI, Hadoop, Dask
Backend & Web:	Spring, RESTful Web Services, JavaScript, JSP, JSF
Operating Systems:	Ubuntu, Microsoft Windows, macOS
Other:	Data management, in-memory indexing & optimization, parallel & distributed systems

Education

PhD, Data Management Jul 2019 – Dec 2024

Department of Computer Science & Engineering, University of Ioannina (UoI) in collaboration with Athena Research Center

- Visiting Researcher (PhD Collaboration), Institute of Computer Science (Data Management research group), Johannes Gutenberg University Mainz, Germany (May 2022 – Jul 2022)

Thesis: In-memory Indexing for Parallel Processing of Single and Multi-Dimensional Queries.

Published papers in: ICDE, VLDBJ, TKDE and SIG-SPATIAL

MSc, Computing Systems: Software and Hardware, Computer Science Nov 2012 – Sep 2016

Department of Informatics and Telecommunications, National and Kapodistrian University of Athens (NKUA)

Thesis: Complex Event Processing (CEP) for Intrusion Detection.

BSc, Computer Science Sep 2006 – Jun 2012

Department of Informatics and Telecommunications, National and Kapodistrian University of Athens (NKUA)

Thesis: Clustering Wikipedia resources.

Additional Experience & Awards

- Best Research Paper Runner-Up — ICDE 2026
- Published papers in top-tier peer-reviewed database conferences and journals (h-index: 5), including SIGMOD, VLDBJ, ICDE and EDBT
- 3rd place — Future of Database Programming Contest, Athens (Mar 2025)
- Program Committee member and external reviewer for 8+ international conferences and journals in data management. Distinguished PC Member, ICDE 2026
- Volunteer: European Data Forum 2014, EDBT/ICDT 2023 Joint Conference, 6th ACM Europe Summer School on Data Science 2025, HDMS 2025

External Links

- LinkedIn: `https://gr.linkedin.com/in/dimitris_tsitsigkos`
- Google Scholar: `https://scholar.google.com/citations?user=z_KVU0QAAAAJ`
- DBLP: `https://dblp.org/pid/163/0537.html`
- GitHub: `https://github.com/dTsitsigkos`
- Personal website: `https://dtsitsigkos.github.io/`

Languages

- Greek (Native)
- English (Fluent)